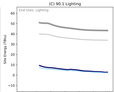
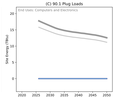
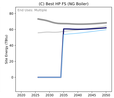
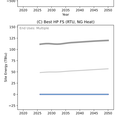
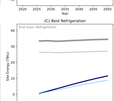

All ECMs
Ref Case Uncongested
w/ Measure(s) Uncongested
Ref Case Compete
w/ Measure(s) Compete
2020
2050

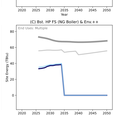

(C) NO.1 Lighting
End Uses: Lighting
Site Energy (TBtu)
Year

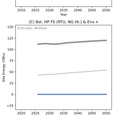

(C) NO.1 Plug Loads
End Uses: Computers and Electronics
Site Energy (TBtu)
Year

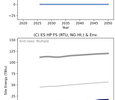

(C) Best HP FS (NG Boiler)
End Uses: Multiple
Site Energy (TBtu)
Year

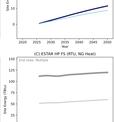

(C) Best HP FS (RTU, NG Heat)
End Uses: Multiple
Site Energy (TBtu)
Year

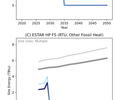

(C) Best Refrigeration
End Uses: Refrigeration
Site Energy (TBtu)
Year

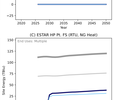

(C) Bst. HP FS (NG Boiler) & Env e x
End Uses: Multiple
Site Energy (TBtu)
Year

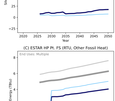

(C) Bst. HP FS (RTU, NG Ht ) & Env e
End Uses: Multiple
Site Energy (TBtu)
Year

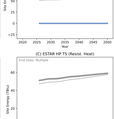

(C) ES HP FS (RTU, NG Ht ) & Env.
End Uses: Multiple
Site Energy (TBtu)
Year

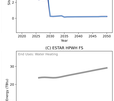

(C) ESTAR HP FS (RTU, NG Heat)
End Uses: Multiple
Site Energy (TBtu)
Year

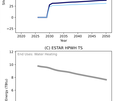

(C) ESTAR HP FS (RTU, Other Fossil Heat)
End Uses: Multiple
Site Energy (TBtu)
Year

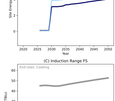

(C) ESTAR HP FS (RTU, NG Heat)
End Uses: Multiple
Site Energy (TBtu)
Year

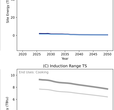

(C) ESTAR HP Ps (RTU, Other Fossil Heat)
End Uses: Multiple
Site Energy (TBtu)
Year

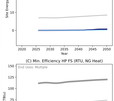

(C) ESTAR HP TS (Resist. Heat)
End Uses: Multiple
Site Energy (TBtu)
Year

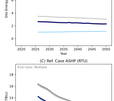

(C) ESTAR HPWH FS
End Uses: Water Heating
Site Energy (TBtu)
Year

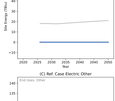

(C) ESTAR HPWH TS
End Uses: Water Heating
Site Energy (TBtu)
Year

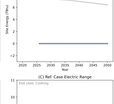

(C) Induction Range FS
End Uses: Cooking
Site Energy (TBtu)
Year

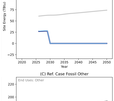

(C) Induction Range TS
End Uses: Cooking
Site Energy (TBtu)
Year

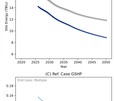

(C) Min. Efficiency HP FS (RTU, NG Heat)
End Uses: Multiple
Site Energy (TBtu)
Year

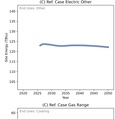

(C) Ref. Case ASHP (RTU)
End Uses: Multiple
Site Energy (TBtu)
Year

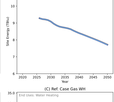

(C) Ref. Case Electric Other
End Uses: Other
Site Energy (TBtu)
Year

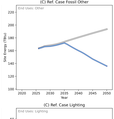

(C) Ref. Case Electric Range
End Uses: Cooking
Site Energy (TBtu)
Year

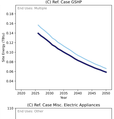

(C) Ref. Case Fossil Other
End Uses: Other
Site Energy (TBtu)
Year

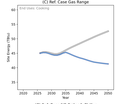

(C) Ref. Case GSHP
End Uses: Heating
Site Energy (TBtu)
Year

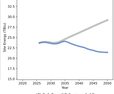

(C) Ref. Case Gas Range
End Uses: Cooking
Site Energy (TBtu)
Year

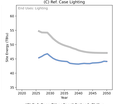

(C) Ref. Case Gas WH
End Uses: Water Heating
Site Energy (TBtu)
Year

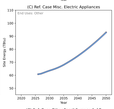

(C) Ref. Case Lighting
End Uses: Lighting
Site Energy (TBtu)
Year

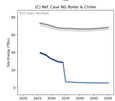

(C) Ref. Case Misc. Electric Appliances
End Uses: Other
Site Energy (TBtu)
Year

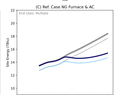

(C) Ref. Case NG Boiler & Chiller
End Uses: Multiple
Site Energy (TBtu)
Year

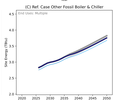

(C) Ref. Case NG Furnace & AC
End Uses: Multiple
Site Energy (TBtu)
Year

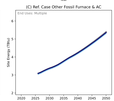

(C) Ref. Case Other Fossil Boiler & Chiller
End Uses: Multiple
Site Energy (TBtu)
Year

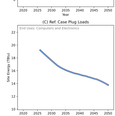

(C) Ref. Case Other Fossil Furnace & AC
End Uses: Multiple
Site Energy (TBtu)
Year

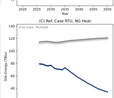

(C) Ref. Case Plug Loads
End Uses: Computers and Electronics
Site Energy (TBtu)
Year

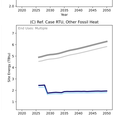

(C) Ref. Case RTU, NG Heat
End Uses: Multiple
Site Energy (TBtu)
Year

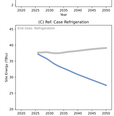

(C) Ref. Case RTU, Other Fossil Heat
End Uses: Multiple
Site Energy (TBtu)
Year

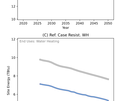

(C) Ref. Case Refrigeration
End Uses: Refrigeration
Site Energy (TBtu)
Year

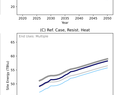

(C) Ref. Case Resist. WH
End Uses: Water Heating
Site Energy (TBtu)
Year

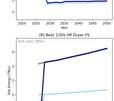

(C) Ref. Case. Resist. Heat
End Uses: Multiple
Site Energy (TBtu)
Year

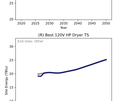

(R) Best 120V HP Dryer FS
End Uses: Other
Site Energy (TBtu)
Year

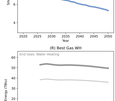

(R) Best 120V HP Dryer TS
End Uses: Other
Site Energy (TBtu)
Year

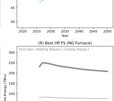

(R) Best Gas WH
End Uses: Water Heating
Site Energy (TBtu)
Year

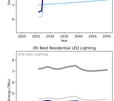

(R) Best HP FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

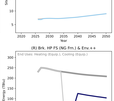

(R) Best Residential LED Lighting
End Uses: Lighting
Site Energy (TBtu)
Year

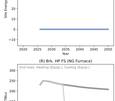

(R) Brk. HP FS (NG Fn.) & Env.++
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

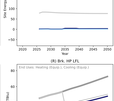

(R) Brk. HP FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

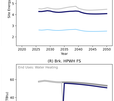

(R) Brk. HP (FL
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

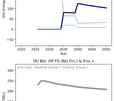

(R) Brk. HPWH FS
End Uses: Water Heating
Site Energy (TBtu)
Year

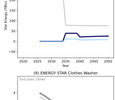

(R) Bst. HP FS (NG Fn.) & Env.+
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

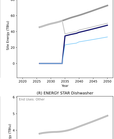

(R) ENERGY STAR Clothes Washer
End Uses: Other
Site Energy (TBtu)
Year

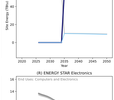

(R) ENERGY STAR Dishwasher
End Uses: Other
Site Energy (TBtu)
Year

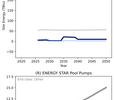

(R) ENERGY STAR Electronics
End Uses: Computers and Electronics
Site Energy (TBtu)
Year

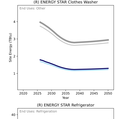

(R) ENERGY STAR Pool Pumps
End Uses: Other
Site Energy (TBtu)
Year

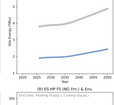

(R) ENERGY STAR Refrigerator
End Uses: Refrigeration
Site Energy (TBtu)
Year

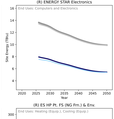

(R) ES HP FS (NG Fn.) & Env.
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

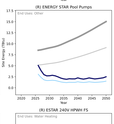

(R) ES HP PS, FS (NG Fn.) & Env.
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

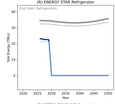

(R) ESTAR 240V HPWH FS
End Uses: Water Heating
Site Energy (TBtu)
Year

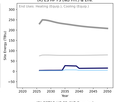

(R) ESTAR GSHP (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

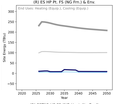

(R) ESTAR HP FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

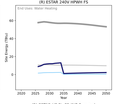

(R) ESTAR HP FS (NG Heat, No Cool)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

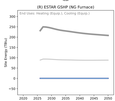

(R) ESTAR HP PS, FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

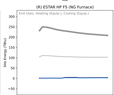

(R) ESTAR HP Ps, FS (NG Heat, No Cool)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year


(R) ESTAR HP TS (Resist.)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

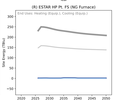

(R) ESTAR HPWH TS
End Uses: Water Heating
Site Energy (TBtu)
Year

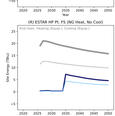

(R) Induction Range FS
End Uses: Cooking
Site Energy (TBtu)
Year

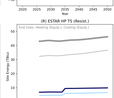

(R) Induction Range TS
End Uses: Cooking
Site Energy (TBtu)
Year

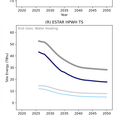

(R) Min. Efficiency ASHP FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

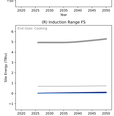

(R) Ref. Case ASHP
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

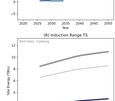

(R) Ref. Case Bst or No Heat & AC
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

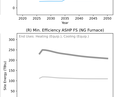

(R) Ref. Case Clothes Washer
End Uses: Other
Site Energy (TBtu)
Year

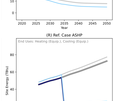

(R) Ref. Case Dishwasher
End Uses: Other
Site Energy (TBtu)
Year

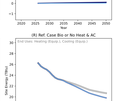

(R) Ref. Case Electric Dryer
End Uses: Other
Site Energy (TBtu)
Year

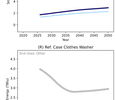

(R) Ref. Case Electric Other
End Uses: Other
Site Energy (TBtu)
Year

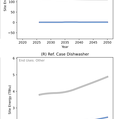

(R) Ref. Case Electric Range
End Uses: Cooking
Site Energy (TBtu)
Year

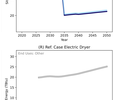

(R) Ref. Case Electronics
End Uses: Computers and Electronics
Site Energy (TBtu)
Year

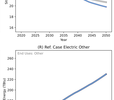

(R) Ref. Case Fans and Pumps
End Uses: Other
Site Energy (TBtu)
Year

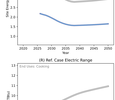

(R) Ref. Case Fossil Other
End Uses: Other
Site Energy (TBtu)
Year

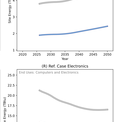

(R) Ref. Case GSHP
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

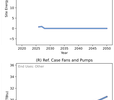

(R) Ref. Case Gas Dryer
End Uses: Other
Site Energy (TBtu)
Year


(R) Ref. Case Gas Range
End Uses: Cooking
Site Energy (TBtu)
Year

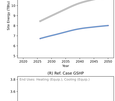

(R) Ref. Case Gas WH
End Uses: Water Heating
Site Energy (TBtu)
Year

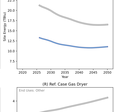

(R) Ref. Case Lighting
End Uses: Lighting
Site Energy (TBtu)
Year

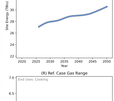

(R) Ref. Case Misc. Electric Appliances
End Uses: Other
Site Energy (TBtu)
Year

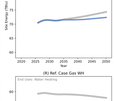

(R) Ref. Case NG Boiler & AC
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

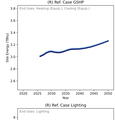

(R) Ref. Case NG Furnace & AC
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

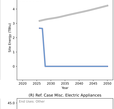

(R) Ref. Case No Cooling
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

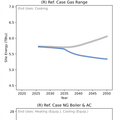

(R) Ref. Case NGHP
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year


(R) Ref. Case Other Fossil Boiler & AC
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year

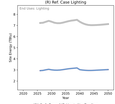

(R) Ref. Case Pool Pumps
End Uses: Other
Site Energy (TBtu)
Year

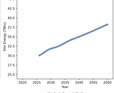

(R) Ref. Case Refrigerator
End Uses: Refrigeration
Site Energy (TBtu)
Year

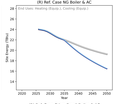

(R) Ref. Case Resist. Heat & AC
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year


(R) Ref. Case Resist. Heat, No Cooling
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year


(R) Ref. Case Resist. WH
End Uses: Water Heating
Site Energy (TBtu)
Year


(R) Ref. NG Furnace & AC & Env.
End Uses: Heating (Equip.), Cooling (Equip.)
Site Energy (TBtu)
Year


(R) Building Codes
End Uses: Multiple
Site Energy (TBtu)
Year


(C) Building Codes
End Uses: Multiple
Site Energy (TBtu)
Year


(R) Building Performance Standards
End Uses: Multiple
Site Energy (TBtu)
Year


(C) Building Performance Standards
End Uses: Multiple
Site Energy (TBtu)
Year